

Jackie (Jiaqi) Zou

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Experience

AI Automation & Full Stack Engineer

May 2026 – Present

Castle Property Management

On-Site, Toronto

- Built internal AI-assisted automation workflows that reduced repetitive leasing and operations tasks by 1000%, accelerating customer-service response times.
- Developed and deployed full-stack internal tools using Python, JavaScript, and modern web frameworks, onboarding 50+ internal users to centralize property data.
- Designed robust backend scripts and automated ETL pipelines for data processing and reporting, eliminating manual data entry by 10 hours per week across business operations.

Data Scientist

Aug 2024 – Present

Depu Consulting Company

Part-time/Remote

- Designed high-throughput pipelines for data cleaning, preprocessing, and storage, successfully managing over 10,000 records of structured and unstructured business data.
- Built advanced data and knowledge-base workflows to help property-management teams seamlessly organize, search, and audit 500+ legal documents and property records.
- Integrated traditional and vector databases (SQL/NoSQL) to enable semantic search and document retrieval, slashing information lookup times for analytics and decision support by 140%.

Projects

Quant Backtesting & Portfolio Optimization Engine

May 2026 – Present

Python, Pandas, NumPy, cvxpy, yfinance, OpenBB, Streamlit, Plotly, Pytest

[GitHub](#)

- Engineered an 11-stage algorithmic trading research engine supporting multi-asset OHLCV data ingestion, modular strategy plug-ins, and interactive dashboard visualizations.
- Implemented rigorous next-bar execution mechanics incorporating commission, slippage, and cash constraints, reducing lookahead bias and matching historical market parameters with 98% accuracy.
- Programmed 2 optimization modules using cvxpy/scipy for mean-variance and minimum-variance optimization, enforcing strict long-only and maximum-weight boundary constraints. with total return: 192.13%
- Developed comprehensive Streamlit and CLI reporting workflows to calculate critical risk-adjusted metrics, including Sharpe, Sortino, Calmar ratios, and maximum drawdown profiles.

CadenSci: AI-Powered Scientific Research Assistant

Jun 2026 – Present

Python, FastAPI, React, TypeScript, Qdrant, PostgreSQL, LangChain, Gemini API, Docker, Cloudflare Tunnel

[GitHub](#)

- Built a production-ready, full-stack RAG research platform combining a FastAPI backend, a React/TypeScript frontend, and a self-hosted Qdrant vector database.
- Automated an ingestion pipeline to download, parse, chunk, and index 1,000+ arXiv papers across CS, math, and quant finance, optimizing chunk sizes to ensure high semantic relevance.
- Designed a hybrid reasoning workflow utilizing vector retrieval and citation grounding to eliminate LLM hallucinations and automatically identify latent academic research gaps.
- Deployed scalable containerized infrastructure via Docker and Cloudflare Tunnels, maintaining public web access while keeping underlying AI data processing securely self-hosted.

Technical Skills

- **Languages:** Python, Java, C/C++, SQL, JavaScript, Go
- **Backend & Databases:** FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, API Design, Query Optimization
- **Frontend & Tools:** React, Next.js, TailwindCSS, HTML/CSS, Git, GitHub, Linux/Unix, Docker, Vercel
- **Data/ML:** Pandas, NumPy, scikit-learn, PyTorch, ETL Pipelines, Data Processing

Education & Awards

University of Waterloo Sep 2025 – Present

Bachelor of Computer Science and Combinatorics & Optimization

Canadian Mathematical Olympiad Qualification 2024 [Honourable Mention](#)

Google IT Support Certificate, Coursera